

Digital Audio Player — Power Budget

Rev 3 | Load analysis, battery sizing and regulator requirements | Alexander Loker

This document converts per-block power figures into three numbers the design needs: an average draw for battery sizing, a peak draw for regulator and thermal sizing, and a standby draw for shelf life. These are different numbers and must not be conflated.

Changes from Rev 2: charger selected (BQ24250RGET), with measured-from-datasheet efficiency, leakage and thermal figures replacing estimates. Standby figure corrected — the charger's high-impedance battery leakage, not SYSOFF, is the relevant path. Charging section rewritten.

1. Method

Summing every block's active figure produces a number no operating state reaches. Two blocks dominate the error:

- **The microSD card is active roughly 2% of the time.** FLAC 16/44.1 stereo is about 90 kB/s; a Class 10 card sustains 10 MB/s. Its 330 mW active figure contributes about 7 mW averaged.
- **The OPA1622's 450 mW is full output into a low-impedance load.** At IEM listening levels the output stage draws little more than quiescent. Quiescent governs battery life; peak governs thermal and regulator sizing.
- **The display is off during playback by design,** putting its duty cycle near 5% in the dominant use case.

2. Source data

Block	Active	Passive	Rail	Confidence
DAC (ES9038Q2M)	40 mW	1.3 mW	3.3 V	Datasheet
MCU (ESP32-S3)	165 mW	7 uW	3.3 V / 1.8 V	Datasheet
microSD (32 GB Class 10)	330 mW	3 mW	3.3 V	Typical
Output amp (OPA1622, 2 ch)	450 mW (max)	5 uA	±3.3 V	Max, not typical
I/V and summing (6 ch)	~110 mW	~10 uW	±3.3 V	Estimated, parts not final
Display + backlight (2.4 in)	300 mW	—	3.3 V + boost	Rough estimate
Charger (BQ24250) leakage	—	~60-80 uW	BAT	Datasheet: 16 uA typ / 22 uA max, high-Z

Note on the op-amp count. The ES9038Q2M's differential current outputs require four I/V channels plus two summing channels in addition to the two OPA1622 output channels — eight total. At roughly 3.6 mA per channel across a 6.6 V span the six non-output channels add about 110 mW of continuous quiescent draw. This is the largest load after the MCU, and it is the price of the current-output architecture.

3. Conversion efficiency

Branch	Topology	Efficiency	Note
Charger input to SYS	BQ24250, 3 MHz sync buck	~90-93%	From the datasheet efficiency curves at 5 V in, 500 mA.
Digital 3.3 V	Buck / buck-boost from SYS	90%	Switcher only.

Branch	Topology	Efficiency	Note
Analog ± 3.3 V	Boost to 5 V, then LT3042 / LT3093	~72%	Boost 90%, LDO 66% on dropout. Deliberate — the LDO loss buys the noise performance.
Backlight	Constant-current boost to LED string	85%	Only when the display is on.

4. Operating scenarios

4.1 S1 — Standby

Block	Draw	Basis
DAC	1.3 mW	Passive
MCU	<0.1 mW	Deep sleep
microSD	0 mW	Power-gated by load switch
Op-amps (8 ch)	<0.1 mW	EN low — shutdown
Display	0 mW	Off
Charger leakage	~0.08 mW	16-22 uA from BAT in high-impedance mode
Regulator quiescent	~5 mW	Estimate; sum actual I_Q for the tree
Total S1	~7 mW	Sets shelf life, not runtime

At 7 mW from a 6.8 Wh cell, standby alone takes roughly 40 days to flatten the battery before cell self-discharge. **This depends on power-gating the SD card and using the OPA1622 EN pin.** Note that **SYSOFF is not the mechanism here** — it would drop charger leakage below 1 uA but disconnects the battery from the system entirely, killing the MCU. It is a shipping mode.

4.2 S2 — Playback, display off (dominant)

Block	Duty	Effective draw	Basis
MCU	100%	165 mW	Decode plus I2S service
DAC	100%	40 mW	Active
microSD	2% / 98%	~10 mW	330 x 0.02 + 3 x 0.98
OPA1622 output (2 ch)	100%	~50 mW	34 mW quiescent + ~16 mW typical signal
I/V and summing (6 ch)	100%	~110 mW	Quiescent
Display	5%	~15 mW	Brief wakes
Load subtotal		390 mW	
Conversion losses		~130 mW	Section 3
Total S2		~520 mW	140 mA at 3.7 V

4.3 S3 — Browsing, display on

Block	Effective draw	Basis
MCU	165 mW	Active

Block	Effective draw	Basis
DAC + full analog chain	200 mW	Not muted during browsing
microSD	~40 mW	Higher duty during directory traversal
Display + backlight	300 mW	Continuous
Conversion losses	~195 mW	
Total S3	~900 mW	243 mA at 3.7 V

4.4 S4 — Absolute peak (transient)

Block	Peak draw	Basis
MCU	165 mW	Active
DAC	40 mW	Active
microSD	330 mW	Read burst
OPA1622 output	450 mW	Maximum output
I/V and summing	130 mW	Quiescent plus signal
Display + backlight	300 mW	Full brightness
Conversion losses	~340 mW	
Total S4	~1755 mW	474 mA at 3.7 V

5. Battery sizing

5.1 Selected cell

Attribute	Value
Manufacturer / MPN	Jauch Quartz — LP605060JU+PCM+2 WIRES 70MM
DigiKey PN	1908-LP605060JU+PCM+2WIRES70MM-ND
Configuration	1S1P Li-Po, 3.7 V nominal, 3.0-4.2 V
Capacity	1850 mAh (6.8 Wh)
Actual envelope	63.0 x 51.0 x 6.6 mm (includes PCM and tab)
Termination	70 mm flying leads — 2 wires, no NTC
Lead time	28 weeks (stock available at time of selection)

5.2 Weighted average and derating

Usage mix 90% S2, 10% S3: $(0.90 \times 520) + (0.10 \times 900) = 558 \text{ mW}$, or **151 mA at 3.7 V**.

Factor	Multiplier	Rationale
Usable depth of discharge	0.90	Cutoff above the PCM threshold so the device shuts down gracefully. The PCM trips around 2.5 V — roughly 20% below where firmware should stop — so relying on it means losing power mid-write.
End-of-life capacity fade	0.85	Li-ion retains ~80-85% of nameplate at 300-500 cycles. Charging at 0.27C rather than 0.5C, and the switching charger's low heat, both favour this figure.
Low-temperature derating	0.95	Device carried outdoors.
Combined	0.73	1850 mAh x 0.73 = 1350 mAh usable

A further **20% design margin** on the load covers the rough display estimate and unverified op-amp quiescent figures: $558 \text{ mW} \times 1.20 = 670 \text{ mW}$, or **181 mA**.

5.3 Result

Condition	Current	Runtime
Weighted average, nameplate capacity	151 mA	12.3 h
Weighted average, derated capacity	151 mA	8.9 h
With 20% design margin, derated capacity	181 mA	7.5 h
Continuous browsing (worst realistic case)	243 mA	5.6 h

The 7 h target is met at 7.5 h under the pessimistic case. The rail reduction to $\pm 2.5 \text{ V}$ and MCU clock reduction to 160 MHz remain **in reserve** — spend them only if the display measures worse than 300 mW.

6. Charging

Parameter	Value	Note
Charge current	500 mA (0.27C)	Bottom of the BQ24250's 500 mA - 2 A range; the datasheet's own design example uses this value.

Parameter	Value	Note
Charge time	~4.5-5 h	Including CV taper.
Safety timer	9 h, set via I2C	The 6 h default leaves under an hour of margin. This is a required configuration step, not an optimisation.
Charger dissipation	~160 mW	At ~93% efficiency from the datasheet curves.
Junction rise	~5 degC	theta-JA 32.9 degC/W (VQFN-24). Regulation threshold 125 degC.
Linear equivalent (rejected)	~650 mW, ~21 degC	Conducted into a cell 2 mm away while it climbs toward full charge. Heat plus high state of charge is the worst case for calendar aging.

- **Input current is not charge current.** Power is conserved through the buck: 500 mA at 5 V in yields roughly 600 mA into a 3.7 V cell at ~90% efficiency. On a port whose current limit cannot be negotiated, this is a real gain over a linear charger's one-for-one ratio.
- **Concurrency is handled in hardware.** PowerPath monitors battery current and reduces charge current when system load exceeds the input limit, so the S4 peak of 474 mA does not collapse the rail or break termination. The battery supplements system load when the adapter cannot meet demand.
- Input current limit set by EN1/EN2 GPIOs (500 mA / 100 mA / external RILIM / Hi-Z) or over I2C.
- VDPM reduces input current to hold the supply up when a weak source or long cable sags. Short to GND for the 4.68 V default.

7. Regulator and thermal sizing

These use S4 (peak), never the average.

- **Charger inductor:** 1.0 uH at 3 MHz, 3.7 V battery from 5 V input, gives roughly 320 mA of ripple. Inductor current is system load plus charge current, since SYS sits after the inductor: worst case 474 + 500 mA plus half the ripple, about **1.15 A**. Specify ≥ 2 A saturation, shielded, low DCR.
- **Analog LDOs:** peak analog branch load is roughly 580 mW across ± 3.3 V. Check LT3042 and LT3093 dissipation at 5 V input — 1.7 V dropout times rail current is heat in the package.
- **Analog boost:** must supply the analog branch at peak plus the backlight. Size inductor saturation against peak.
- **Load-step response:** SD read bursts and output-stage transients are step loads on their rails. The charger's own loop switches from voltage-mode (Type-3 compensated) in CC to current-mode in CV, so SYS transient behaviour differs between charge phases — expect this during bring-up.
- **OPA1622 dissipation:** worst-case class AB dissipation occurs near $0.64 \times V_{\text{supply}}$ peak output, which at ± 3.3 V is 2.1 V peak — above the 1.41 V design maximum, so the device sits on the favourable side of the curve.
- **SYS and BAT absolute maximum is 5 V.** Anything downstream must respect this.

8. Confidence register

Item	Status	Action
Display + backlight (300 mW)	Rough estimate	Measure the selected 2.4 inch panel at operating brightness. Largest single uncertainty.
OPA1622 quiescent at ± 3.3 V	Not yet read	Read from the Quiescent Current vs. Supply Voltage curve at 6.6 V total.
OPA1622 shutdown current	Not yet read	Needed for the S1 standby figure.
I/V and summing op-amp selection	Not chosen	Quiescent current is a selection criterion — six channels at stake.
ES9038Q2M (40 mW)	Datasheet	Confirm which power mode and oversampling setting applies.

Item	Status	Action
microSD figures	Typical	Verify against an industrial card datasheet or measure with a sense resistor in the card's ground return. Datasheet figures are 1-second averages and hide peaks.
SD duty cycle (2%)	Calculated	Validate by instrumenting buffer refill frequency.
Cell capacity (1850 mAh)	Datasheet	Verify the discharge rate the figure is quoted at; the design draws $\sim 0.08C$, well below the typical $0.2C$ test rate.
Charger efficiency ($\sim 93\%$)	Datasheet curve	Read at 5 V input, 500 mA output. Verify against measurement at the actual operating point.
Conversion efficiencies (other branches)	Assumed	Replace with LTspice results.
Regulator quiescent (5 mW)	Estimate	Sum actual I_Q for every regulator in the tree.

9. Validation plan

- Instrument battery current with a sense resistor in the cell return, logged over a full discharge. This supersedes every estimate here.
- Log firmware state alongside current so scenarios can be separated rather than inferred.
- Characterise the spare cell's discharge curve at operating current — required anyway to build the voltage-based fuel gauge lookup table, so it serves two purposes.
- Measure a full charge cycle: current profile, time to termination, charger and cell temperature. Confirm the 9 h safety timer is never approached and that thermal regulation never engages.
- Verify the TS network's actual trip temperatures rather than trusting the calculated values — beta error alone shifts the cool and warm crossings by about 3 degC.
- Measure each branch independently at Rev A, while branches are still physically accessible.
- Re-issue as Rev 4 once the display is measured and op-amp selections are final.

Rev 3. Section 5.3 remains the operative finding: 7.5 h against a 7 h target, with the display estimate as the dominant remaining uncertainty.